Lab-5: Cascading Style Sheets (CSS) @import CSS, Apply style to ordered List, Descendant Selectors, child selectors, first-line, first-letter, :before, :after pseudo element, positioning, BOX Model.

Code:

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Mastering Web Technologies</title>
   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
   <link rel="stylesheet" href="style.css">
</head>

<body>

   <header class="main-header">
      <div class="container-fluid">
         <h1>Web Development Masterclass</h1>
```

```
    </div>
  </header>

  <nav class="left-menu">
    <ul>
      <li><a href="#toc">Table of Content</a></li>
      <li><a href="#layout">Page Layout</a></li>
      <li><a href="#box-model">Box Model</a></li>
      <li><a href="#css3-bootstrap">CSS3 & Bootstrap</a></li>
      <li><a href="#bouncing-ball">Bouncing Ball</a></li>
      <li><a href="#bicycle-anim">Moving Bicycle</a></li>
    </ul>
  </nav>

  <main class="main-content">
    <div class="container">

      <section id="toc" class="mb-5">
        <h2>Module 1: Course Syllabus</h2>
        <div class="toc-container">
          <h3>Course: Web Technologies</h3>
          <h4>Table of Contents</h4>

          <ol class="toc-root">
            <li>HTML5 Structure & Semantics</li>
            <li>CSS3 Styling & Design
```

```
            <ol>
              <li>Introduction to Cascading Style Sheets</li>
              <li>Integration Methods
                <ol>
                  <li>External Stylesheets
                    <ol>
                      <li>Best Practices for Scalability</li>
                    </ol>
                  </li>
                </ol>
              </li>
              <li>Cross-Browser Compatibility</li>
            </ol>
          </li>
          <li>XML & Data Representation</li>
        </ol>
      </div>
    </section>

    <hr>

    <section id="layout" class="mb-5">
      <h2>Module 2: Layout & Positioning Patterns</h2>
      <p>
        Understanding how to structure web applications using advanced positioning techniques.
```

```
        </p>
        <ul>
          <li><strong>Sticky Header:</strong> Keeps navigation accessible while
scrolling.</li>
          <li><strong>Fixed Sidebar:</strong> Persistent access to course modules.</li>
          <li><strong>Fluid Content:</strong> Adapts to remaining screen real estate.</li>
        </ul>
        <div class="position-demo relative-box">
          <strong>Feature Highlight</strong><br>
          (Relative Positioning Flow)
          <div class="absolute-box">
            <strong>New!</strong><br>
            (Absolute Overlay)
          </div>
        </div>
      </section>

      <hr>

      <section id="box-model" class="mb-5">
        <h2>Module 3: The CSS Box Model</h2>
        <p>Visualizing the fundamental architecture of every DOM element.</p>

        <div class="box-model-container">
          <div class="margin-area">
            Margin (External Spacing)
```

```
            <div class="border-area">
              Border (Boundary)
              <div class="padding-area">
                Padding (Internal Spacing)
                <div class="content-area">
                  Core Content
                </div>
              </div>
            </div>
          </div>
        </div>
      </section>

      <hr>

      <section id="css3-bootstrap" class="mb-5">
        <h2>Module 4: Modern UI Components</h2>
        <p>Leveraging frameworks and modern CSS features for rapid development.</p>

        <div class="row">
          <div class="col-md-6">
            <div class="card shadow-sm border-primary">
              <div class="card-header bg-primary text-white">
                Featured Course
              </div>
              <div class="card-body">
```

```
                <h5 class="card-title">Full-Stack Development</h5>
                <p class="card-text">Master the MERN stack with our comprehensive guide. Build
                  real-world applications including social media platforms, dashboards, and e-commerce
                  sites.</p>
                <button class="btn btn-primary">Enroll Now</button>
              </div>
            </div>
          </div>

          <div class="col-md-6">
            <div class="css3-demo-box">
              <strong>Interactive Element</strong><br>
              Hover for 3D Transform
            </div>
            <div class="css3-animation-box mt-3">
              <strong>Motion UI</strong><br>
              CSS3 Keyframe Animation
            </div>
          </div>
        </div>
      </section>

      <hr>
```

```
    <section id="bouncing-ball" class="mb-5">
      <h2>Module 5: Physics Simulation</h2>
      <p>Creating realistic motion effects without JavaScript.</p>

      <div class="bouncing-container">
        <div class="ball"></div>
      </div>
    </section>

    <hr>

    <section id="bicycle-anim" class="mb-5">
      <h2>Module 6: Moving Bicycle</h2>
      <p>Simple CSS Animation of a bicycle running in a rectangle.</p>

      <div class="bicycle-container">
        <div class="bicycle">
          <div class="wheel front"></div>
          <div class="wheel back"></div>
          <div class="frame"></div>
          <div class="seat"></div>
          <div class="handle"></div>
        </div>
      </div>
    </section>
  </div>
```

```
  </main>


  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></sc
ript>
</body>


</html>
```

Style.css

```
body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  margin: 0;
  padding: 0;
  background-color: #f4f4f4;
}

.main-header {
  background-color: #343a40;
  color: white;
  padding: 1rem;
  text-align: center;
  position: sticky;
  top: 0;
```

```
    z-index: 1000;
    box-shadow: 0 2px 5px rgba(0, 0, 0, 0.2);
}

.left-menu {
    background-color: #2c3e50;
    color: white;
    width: 200px;
    height: 100vh;
    position: fixed;
    top: 60px;
    left: 0;
    padding-top: 20px;
    z-index: 900;
}

.left-menu ul {
    list-style-type: none;
    padding: 0;
}

.left-menu li a {
    display: block;
    color: white;
    padding: 10px 20px;
    text-decoration: none;
```

```
    transition: background 0.3s;
}

.left-menu li a:hover {
    background-color: #34495e;
}

.main-content {
    margin-left: 200px;
    padding: 20px;
    min-height: 200vh;
    background-color: #fff;
    position: relative;
    z-index: 1;
}

.position-demo {
    height: 200px;
    border: 1px solid #ccc;
    margin-top: 20px;
    background-color: #e9ecef;
}

.relative-box {
    position: relative;
    top: 20px;
```

```
    left: 20px;
    background-color: #d1ecf1;
    padding: 20px;
    width: 300px;
    border: 1px solid #bee5eb;
    z-index: 10;
}

.absolute-box {
    position: absolute;
    top: 50px;
    right: -50px;
    width: 150px;
    height: 100px;
    background-color: #f8d7da;
    border: 1px solid #f5c6cb;
    display: flex;
    align-items: center;
    justify-content: center;
    z-index: 20;
    box-shadow: 3px 3px 5px rgba(0, 0, 0, 0.2);
}

.toc-container {
    background: white;
    padding: 20px;
```

```
    border: 1px solid #ddd;
    display: inline-block;
    min-width: 400px;
    box-shadow: 2px 2px 10px rgba(0, 0, 0, 0.1);
}

ol.toc-root {
    counter-reset: chapter;
    list-style-type: none;
    padding-left: 0;
}

ol.toc-root>li {
    counter-increment: chapter;
    margin-bottom: 5px;
    font-weight: bold;
}

ol.toc-root>li:before {
    content: "Chapter " counter(chapter) ". ";
    font-weight: bold;
    color: #333;
}

ol.toc-root>li>ol {
    counter-reset: sub-section;
```

```
    list-style-type: none;
    margin-top: 5px;
}

ol.toc-root>li>ol>li {
    counter-increment: sub-section;
    font-weight: normal;
    margin-left: 20px;
}

ol.toc-root>li>ol>li:before {
    content: counter(chapter) "." counter(sub-section) " ";
    color: #555;
    margin-right: 5px;
}

ol.toc-root>li>ol>li>ol {
    counter-reset: sub-sub-section;
    list-style-type: none;
    margin-top: 5px;
}

ol.toc-root>li>ol>li>ol>li {
    counter-increment: sub-sub-section;
    margin-left: 20px;
}
```

```
ol.toc-root>li>ol>li>ol>li:before {
   content: counter(chapter) "." counter(sub-section) "." counter(sub-sub-section) " ";
   color: #777;
   margin-right: 5px;
}

ol.toc-root>li>ol>li>ol>li>ol {
   counter-reset: deep-section;
   list-style-type: none;
   margin-top: 5px;
}

ol.toc-root>li>ol>li>ol>li>ol>li {
   counter-increment: deep-section;
   margin-left: 20px;
}

ol.toc-root>li>ol>li>ol>li>ol>li:before {
   content: counter(chapter) "." counter(sub-section) "." counter(sub-sub-section) "."
counter(deep-section) " ";
   font-style: italic;
   margin-right: 5px;
}

.box-model-container {
```

```
    padding: 20px;
    background-color: #fff;
    border: 1px solid #ddd;
    text-align: center;
    font-family: monospace;
}

.margin-area {
    background-color: #f8d7da;
    padding: 40px;
    border: 1px dashed #dc3545;
    display: inline-block;
    position: relative;
}

.border-area {
    background-color: #ffeeba;
    padding: 40px;
    border: 5px solid #ffc107;
    display: inline-block;
}

.padding-area {
    background-color: #d4edda;
    padding: 40px;
    display: inline-block;
```

```
}

.content-area {
   background-color: #cce5ff;
   width: 200px;
   height: 100px;
   line-height: 100px;
   display: inline-block;
   border: 1px solid #b8daff;
   font-weight: bold;
}

.css3-demo-box {
   background: linear-gradient(45deg, #667eea, #764ba2);
   color: white;
   padding: 20px;
   border-radius: 10px;
   text-align: center;
   cursor: pointer;
   transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.css3-demo-box:hover {
   transform: scale(1.05) rotate(2deg);
   box-shadow: 0 10px 20px rgba(0, 0, 0, 0.2);
}
```

```
@keyframes slideIn {
  0% {
    transform: translateX(-100%);
    opacity: 0;
  }

  100% {
    transform: translateX(0);
    opacity: 1;
  }
}

.css3-animation-box {
  background-color: #20c997;
  color: white;
  padding: 15px;
  text-align: center;
  border-radius: 5px;
  animation: slideIn 2s ease-out infinite alternate;
}

.bouncing-container {
  width: 300px;
  height: 400px;
  border: 5px solid #333;
```

```
  position: relative;
  background-color: #f0f0f0;
  margin: 0 auto;
}

.ball {
  width: 50px;
  height: 50px;
  background-color: red;
  border-radius: 50%;
  position: absolute;
  bottom: 0;
  left: 50%;
  transform: translateX(-50%);
  animation: simpleBounce 2s infinite ease-in-out;
}

@keyframes simpleBounce {

  0%,
  100% {
    bottom: 0;
  }

  50% {
    bottom: 350px;
```

```
    }
}



.bicycle-container {
    width: 100%;
    height: 150px;
    background-color: #e0f7fa;
    /* Light blue background */
    border: 2px solid #0288d1;
    position: relative;
    overflow: hidden;
}

.bicycle {
    width: 100px;
    height: 60px;
    position: absolute;
    bottom: 20px;
    left: -100px;
    /* Start outside the left edge */
    animation: moveRight 5s linear infinite;
}

.wheel {
    width: 40px;
```

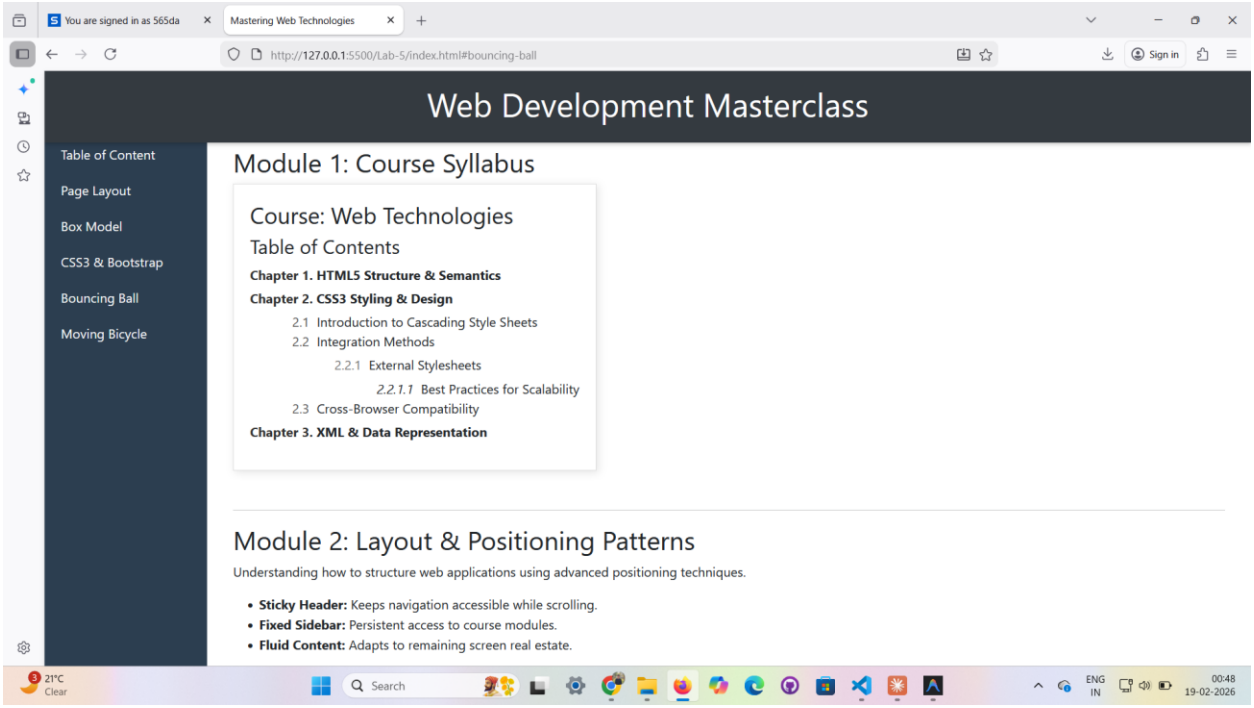
```
  height: 40px;
  border: 3px solid #01579b;
  /* Dark blue wheels */
  border-radius: 50%;
  position: absolute;
  bottom: 0;
  /* Creating simple spokes with gradient */
  background: linear-gradient(0deg, transparent 48%, #01579b 48%, #01579b 52%, transparent 52%),
    linear-gradient(90deg, transparent 48%, #01579b 48%, #01579b 52%, transparent 52%);
  animation: spin 1s linear infinite;
}

.front {
  right: 0;
}

.back {
  left: 0;
}

.frame {
  position: absolute;
  bottom: 20px;
  left: 20px;
```

```
    width: 60px;
    height: 3px;
    background-color: #01579b;
    /* Simple frame bar connecting wheels */
}

/* Simple handle/seat representation */
.frame::before {
    /* Seat post */
    content: '';
    position: absolute;
    left: 10px;
    bottom: 0;
    width: 3px;
    height: 25px;
    background-color: #01579b;
}

.handle {
    /* Handlebar post */
    position: absolute;
    right: 20px;
    bottom: 20px;
    width: 3px;
    height: 30px;
    background-color: #01579b;
```

```
    transform: rotate(-15deg);
}

.handle::after {
    /* Handlebar top */
    content: '';
    position: absolute;
    top: 0;
    left: -5px;
    width: 15px;
    height: 3px;
    background-color: #01579b;
}

@keyframes moveRight {
    0% {
        left: -120px;
    }

    100% {
        left: 100%;
    }
}

@keyframes spin {
    0% {
```

```
      transform: rotate(0deg);
    }

    100% {
      transform: rotate(360deg);
    }
}
```

Output:


You are signed in as 565da
Mastering Web Technologies
http://127.0.0.1:5500/Lab-5/index.html#bouncing-ball
Web Development Masterclass
Table of Content
Page Layout
Box Model
CSS3 & Bootstrap
Bouncing Ball
Moving Bicycle
Module 1: Course Syllabus
Course: Web Technologies
Table of Contents
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Chapter 2. CSS3 Styling & Design
2.1 Introduction to Cascading Style Sheets
2.2 Integration Methods
2.2.1 External Stylesheets
2.2.1.1 Best Practices for Scalability
2.3 Cross-Browser Compatibility
Chapter 3. XML & Data Representation
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Understanding how to structure web applications using advanced positioning techniques.
Sticky Header: Keeps navigation accessible while scrolling.
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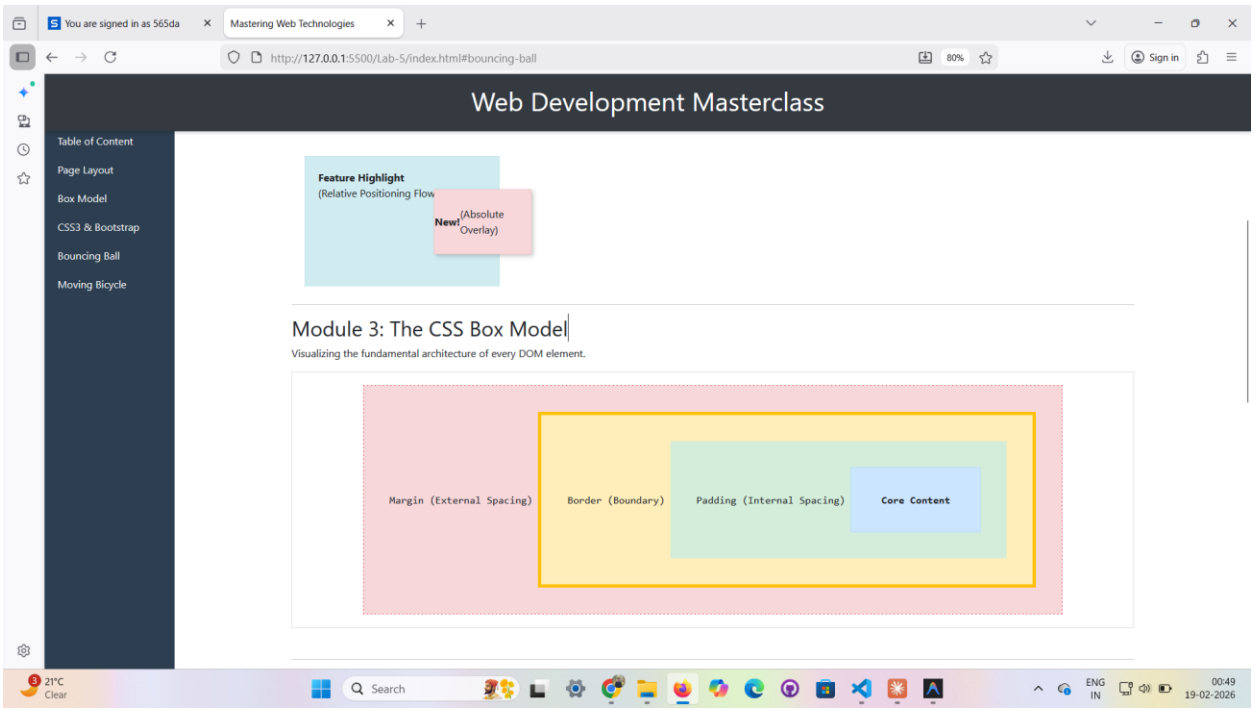
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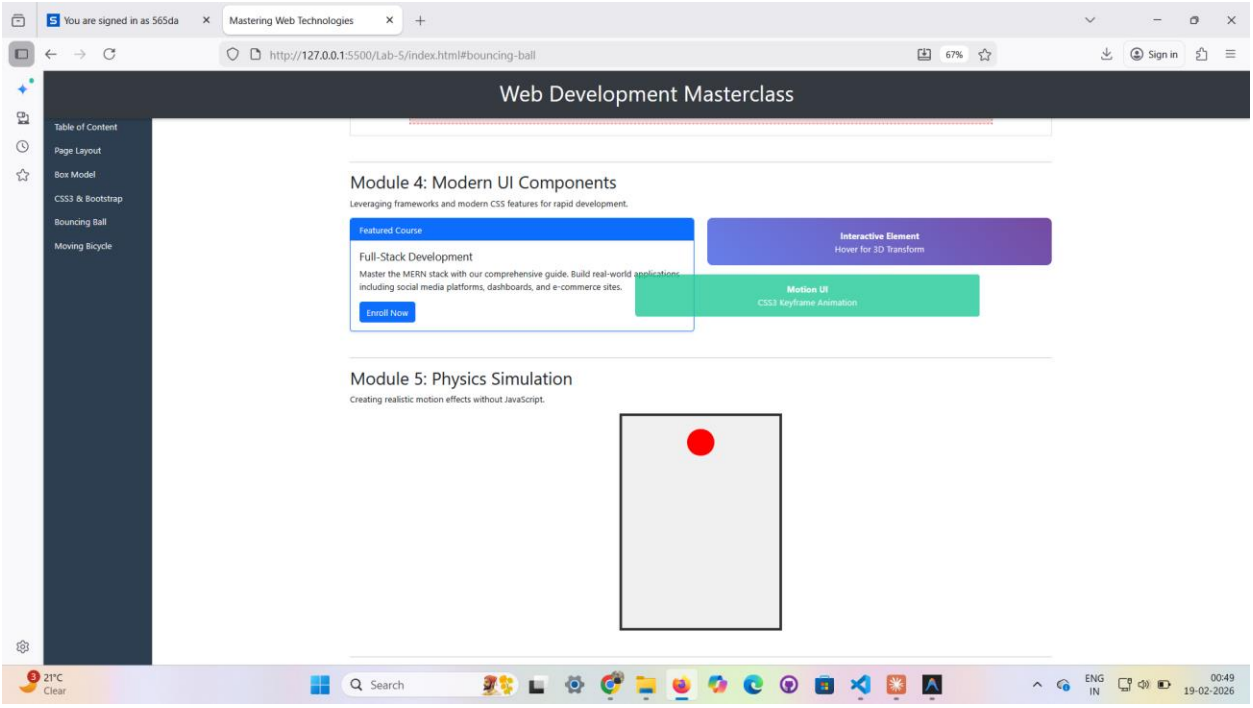
Border (Boundary)
Padding (Internal Spacing)
Core Content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Interactive Element
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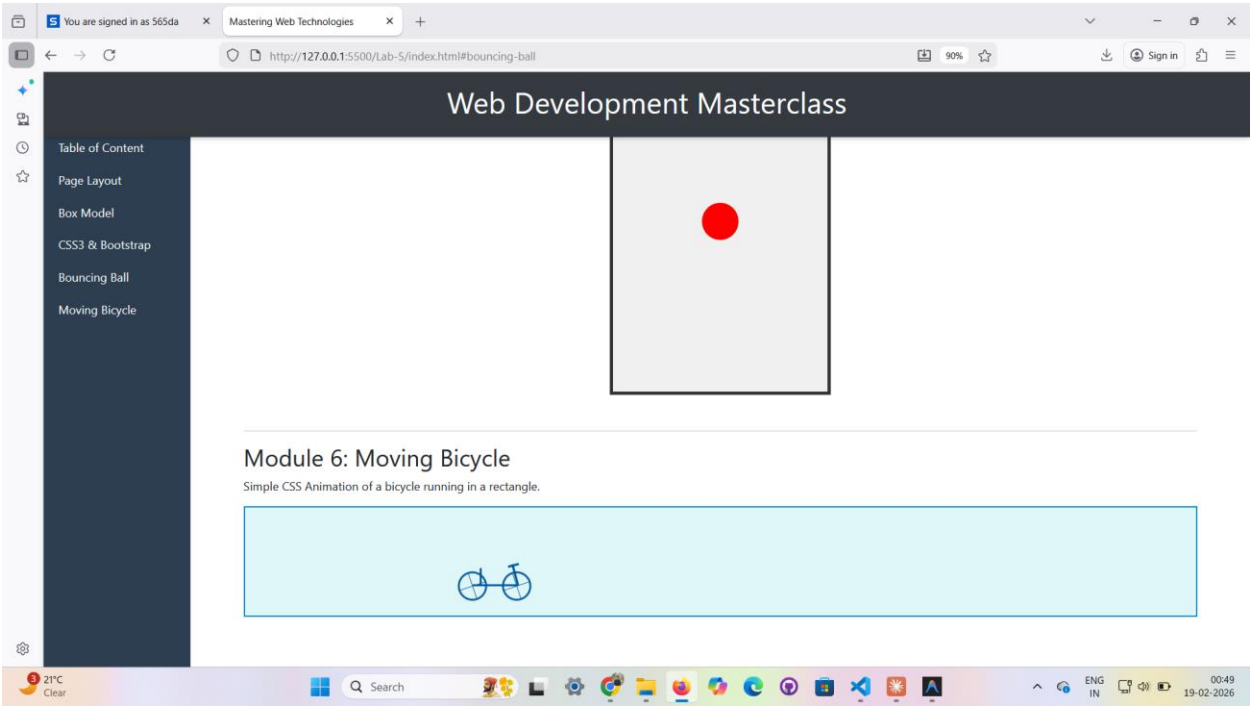
Creating realistic motion effects without JavaScript.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Module 6: Moving Bicycle
Simple CSS Animation of a bicycle running in a rectangle.